PROJECT DOCUMENTATION

Polynomial Regression for School Attendance

Python • Pandas • Scikit-learn • Matplotlib • Seaborn

1. Project Overview

This notebook builds a Polynomial Regression model to examine the relationship between the number of students enrolled and the number of students present. The model uses the Enrolled column as the input feature and the Present column as the target.

2. Project Objectives

- Load and inspect school daily-attendance data from a CSV file.
- Use Enrolled students to predict Present students.
- Train a degree-2 Polynomial Regression model.
- Evaluate predictions using Mean Squared Error (MSE) and R^2 score.
- Plot actual attendance observations and the fitted regression curve.

3. Dataset

Dataset file: 2018-2019_Daily_Attendance_20240429.csv

The notebook loads the file from a Kaggle input path for the school-attendance dataset. The code specifically expects these columns:

- Enrolled — number of students enrolled (input feature, X).
- Present — number of students present (target value, y).

Note: the notebook does not document the dataset's full schema, row count, date coverage, or missing-value checks. Those should be verified before drawing conclusions.

4. Libraries Used

- **Pandas:** Reads the CSV and stores the data in a DataFrame.
- **NumPy:** Creates arrays, random samples, and values for the regression curve.
- **Matplotlib:** Displays the scatter plot and fitted curve.
- **Seaborn:** Imported, but not used in the current notebook cells.
- **Scikit-learn:** Splits the data, creates polynomial features, trains the model, and evaluates it.

5. Methodology / Workflow

1. **Import libraries —** Import the Python packages required for data handling, visualization, model training, and evaluation.
2. **Load data —** Read the CSV file into a Pandas DataFrame and display the first rows with df.head().
3. **Select feature and target —** Set X to the Enrolled column and y to the Present column.
4. **Split the dataset —** Use train_test_split with 20% of the rows reserved for testing and random_state=42 for a repeatable split.
5. **Create polynomial features —** Set degree=2 and transform the input so the model can fit a quadratic relationship.
6. **Train the model —** Fit LinearRegression on the transformed training data.
7. **Predict and evaluate —** Predict the test target values and calculate MSE and R^2.
8. **Visualize —** Plot a sample of actual observations and the model's predicted polynomial curve.

6. Model Details

Algorithm: Polynomial Regression implemented using PolynomialFeatures and LinearRegression.

Polynomial degree: 2 (quadratic).

Training/testing split: 80% training and 20% testing.

Conceptually, the model fits a curve of the form $y = b_0 + b_1x + b_2x^2$, where x is the number of enrolled students and y is the predicted number of present students.

7. Evaluation Metrics

Mean Squared Error (MSE): measures the average squared difference between actual and predicted attendance values. Lower is generally better.

R^2 score: indicates how well the model explains variation in the target values. A value closer to 1 generally indicates a better fit; the score should be interpreted alongside the data and MSE.

The notebook prints the actual MSE and R^2 values when run. Those numeric results are not embedded in the notebook source available for this documentation, so no scores are claimed here.

8. Visualization

The chart is titled “Polynomial Regression: Enrolled vs Present Students.” Blue points represent sampled actual observations, while the red line represents the degree-2 polynomial model's predictions over a range of enrollment values. The axes are labeled Enrolled Students and Present Students.

9. Important Issue to Fix

Potential plotting error: The code selects exactly 1,000 test rows with np.random.choice(..., size=1000, replace=False). If the test set contains fewer than 1,000 rows, this line raises a ValueError. A safer replacement is:

```
sample_size = min(1000, len(X_test))
indices = np.random.choice(len(X_test), size=sample_size, replace=False)
```

Also check for missing values and confirm that Enrolled and Present are numeric. The plot's random sample is not fixed with a random seed, so the displayed blue points can change between runs.

10. How to Run the Notebook

9. Open the notebook in Kaggle or another Python environment with the required libraries installed.
10. Make sure the CSV dataset path exists in that environment; update file_path if needed.
11. Run the cells from top to bottom.
12. Review the printed MSE and R^2 score, then inspect the plot.
13. If running locally, install the dependencies as needed: pandas, numpy, matplotlib, seaborn, and scikit-learn.

11. Conclusion

This project demonstrates a basic machine-learning workflow using school attendance data: loading data, choosing a feature and target, splitting the data, generating polynomial features, fitting a regression model, evaluating it, and visualizing the relationship. The model's usefulness should be judged from the actual evaluation scores and data quality checks. A relationship between enrollment and attendance alone does not establish why attendance changes or guarantee accurate predictions for future dates.

12. Future Improvements

- Add data cleaning and missing-value checks.
- Compare degree 1, 2, and higher-degree models using a consistent evaluation process.
- Use cross-validation and inspect residuals to check model behavior.
- Include additional useful features, such as date or school, if they are present and appropriate in the dataset.
- Save the model and document the measured MSE and R^2 from a successful run.